230507: metadacite, Junior Hill prospect

(Little Peters Formation, Gindalbie Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Leonora (SH 51-1), Weebo (3141)
MGA Zone 51, 310325E 6865442N

Sampled on 12 July 2018

This sample was collected from the 229.58 – 230.79 m depth interval of drillcore JHDD00014, drilled in 2008 by St Barbara Limited at their Junior Hill prospect (Mukherji, 2009). The drillhole is located on Weebo Station and northwest of the Teutonic mining centre, about 7.2 km west-southwest of Garden Well, 6.8 northwest of Junior Hill, and 4.3 km northeast of Clifford Bore.

Geological context

The unit sampled is a metadacite (Fig. 1) assigned to the Little Peters Formation of the 2694–2676 Ma Gindalbie Group (Wyche et al., 2016, and references within). The Little Peters Formation consists mainly of felsic metavolcanic and metavolcaniclastic rocks interbedded with metasandstone and shale (Stewart, 2004; Sapkota and De Paoli, 2016). The formation overlies the Kents Bore Basalt and is tectonically juxtaposed against mafic and ultramafic rocks of the Marshall Pool Subgroup and metasedimentary rocks of the Mount Leonora Formation (Wyche et al., 2016). The present sample was collected to determine the age of fragmental metarhyolite above black shale. A metarhyolite from about 336 m depth in this drillcore yielded an igneous crystallization age of 2660 ± 6 Ma (GSWA 230514, Wingate et al., 2022). Zircons from a felsic igneous rock about 5.7 km to the east-southeast in drillcore JHDD00009 yielded a SHRIMP U–Pb date of 2692 ± 7 Ma (sample LNN009794, Mukherji, 2009). A porphyritic metadacite of the underlying Teutonic Bore Formation, sampled about 13.7 km to the south-southeast, yielded an igneous crystallization age of 2692 ± 4 Ma (GSWA 112159, Nelson, 1995). Pidgeon and Wilde (1990) reported an isotope dilution thermal ionization mass spectrometric (IDTIMS) U–Pb upper intercept age of 2688 ± 4 Ma for multigrain zircon fractions from an outcrop sample of Teutonic Bore Formation rhyolite about 10.5 km to the southeast.

Petrographic description

The sample is an altered, coarse-grained, plagioclase-phyric metadacite (Fig. 1), consisting of about 65–70% groundmass, 25–30% plagioclase phenocrysts, 4–5% pseudomorphed ferromagnesian phenocrysts, <1% quartz phenocrysts, <1% pyrite, and accessory zircon. The groundmass is a microcrystalline quartzofeldspathic aggregate on which is superimposed a phyllic alteration assemblage of carbonate–sericite–quartz–pyrite. Carbonate minerals are abundant and form elongate grains up to 1 mm long that link fragments of fractured plagioclase phenocrysts and occur as finer grains in irregular aggregates and disseminated throughout the groundmass. Sericite forms folia in the groundmass that parallel folia of iron oxide minerals. Plagioclase forms subhedral to euhedral, tabular phenocrysts up to 8 mm long that are albitized and fractured. Some elongate plagioclase phenocrysts are moderately well aligned. Quartz phenocrysts are mostly anhedral, up to 1 mm in diameter, round to strongly embayed, and exhibit weak to moderate undulose extinction. Less abundant secondary quartz forms fibrous grains, together with carbonate, filling fractures within plagioclase phenocrysts. Disseminated subhedral to euhedral pyrite is up to 0.5 mm in size. Pseudomorphed ferromagnesian phenocrysts were probably biotite, and are replaced by chlorite, carbonate and fine iron oxide minerals. The best-preserved grains are tabular, up to 2 mm long, retain a lamellar texture, and most are aligned, and a few are kinked. The sample has been deformed, probably at low metamorphic grade (lower greenschist facies), and has a weak to moderate anastomosing foliation, best defined by phenocryst alignment and sericite–iron oxide folia in the groundmass.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown or opaque, and subhedral to euhedral. The crystals are up to 250 µm long, and mainly elongate, with aspect ratios up to 5:1. Concentric zoning is ubiquitous, many zircons contain high-U, metamict zones, and some appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 11–12 June 2020, using SHRIMP-B. Nine analyses of the M257 standard were obtained during the session, of which six indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.20% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.22%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Fifteen analyses were obtained from 15 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

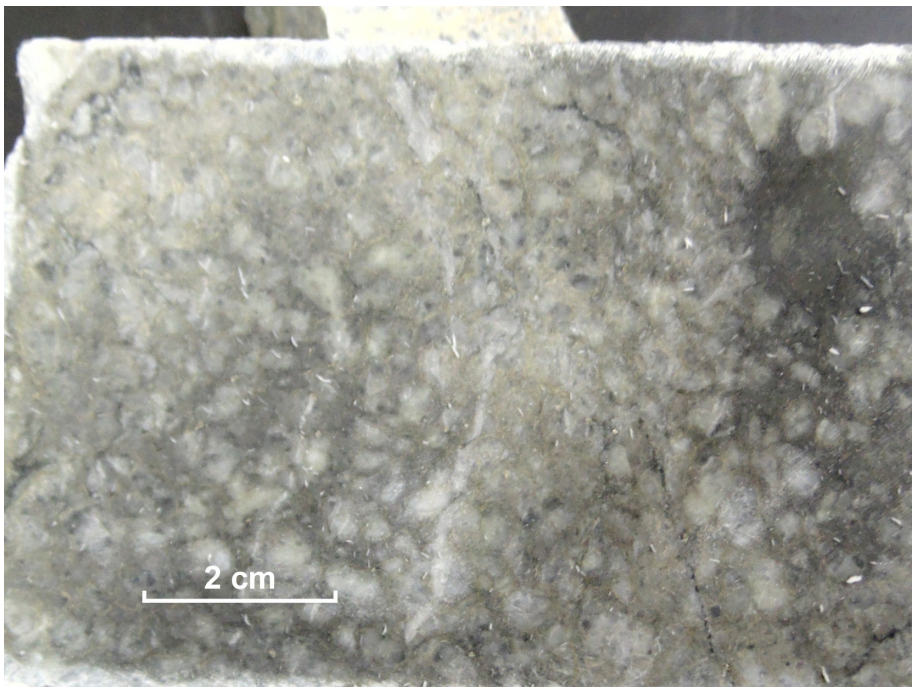

Figure 1. Drillcore image of sample 230507: metadacite, Junior Hill prospect

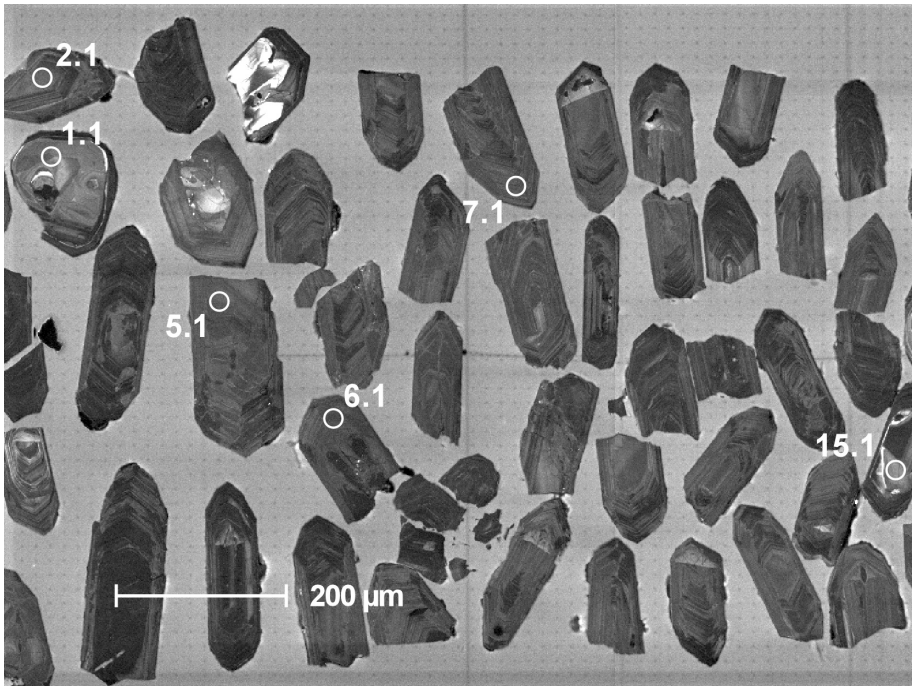

Figure 2. Cathodoluminescence image of representative zircons from sample 230507: metadacite, Junior Hill prospect. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 230507: metadacite, Junior Hill prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	8	8.1	462	439	0.95	0.036	1.989	0.016	0.17931	0.00064	1.989	0.016	0.17898	0.00065	2625	17	2643	6	0.7
I	3	3.1	417	96	0.23	0.126	2.063	0.017	0.18022	0.00068	2.066	0.017	0.17910	0.00072	2545	17	2645	7	3.8
I	9	9.1	436	85	0.20	0.027	2.045	0.016	0.17970	0.00093	2.046	0.016	0.17946	0.00093	2566	17	2648	9	3.1
I	15	15.1	130	9	0.07	0.091	2.044	0.022	0.18027	0.00105	2.046	0.022	0.17947	0.00110	2566	23	2648	10	3.1
I	4	4.1	418	79	0.19	-0.004	2.002	0.016	0.18018	0.00068	2.002	0.016	0.18022	0.00068	2612	18	2655	6	1.6
I	5	5.1	258	4	0.02	-0.014	1.987	0.018	0.18078	0.00082	1.987	0.018	0.18090	0.00082	2628	20	2661	8	1.2
P	11	11.1	546	106	0.19	0.074	2.201	0.017	0.16568	0.00061	2.203	0.017	0.16502	0.00063	2413	16	2508	6	3.8
P	7	7.1	582	144	0.25	0.077	2.184	0.017	0.16776	0.00064	2.185	0.017	0.16707	0.00065	2429	16	2529	7	3.9
P	14	14.1	409	32	0.08	0.067	2.157	0.017	0.16943	0.00066	2.159	0.017	0.16883	0.00067	2454	16	2546	7	3.6
P	2	2.1	644	244	0.38	0.041	2.141	0.018	0.17001	0.00388	2.142	0.018	0.16965	0.00388	2470	17	2554	38	3.3
P	10	10.1	476	59	0.12	0.028	2.063	0.016	0.17074	0.00421	2.064	0.016	0.17048	0.00421	2547	17	2562	41	0.6
X	13	13.1	243	128	0.53	0.061	1.963	0.018	0.18472	0.00079	1.964	0.018	0.18418	0.00081	2653	20	2691	7	1.4
X	1	1.1	76	30	0.40	0.133	1.873	0.025	0.19087	0.00137	1.876	0.025	0.18968	0.00146	2755	31	2739	13	-0.6
D	6	6.1	428	115	0.27	0.128	2.190	0.018	0.17598	0.00100	2.193	0.018	0.17485	0.00103	2422	17	2605	10	7.0
D	12	12.1	423	196	0.46	0.029	2.324	0.019	0.16758	0.00070	2.325	0.019	0.16732	0.00071	2307	16	2531	7	8.9

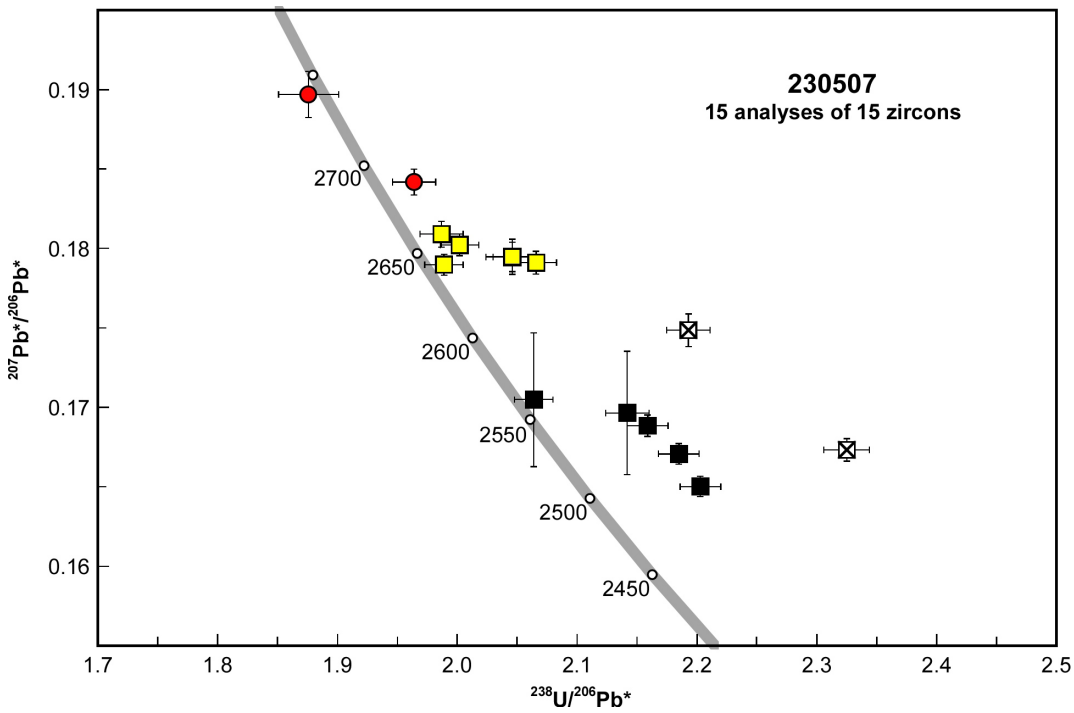

Figure 3. **U–Pb analytical data for sample 230507: metadacite, Junior Hill prospect. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircon cores); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)**

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and considered not to be geologically significant. The remaining 13 analyses can be divided into three groups, based on their ^{207}Pb*/^{206}Pb* ratios and their positions within the crystals.

Group I comprises six analyses (Table 1), which yield a weighted mean ^{207}Pb*/^{206}Pb* date of 2650 ± 6 Ma (MSWD = 0.45).

Group X comprises two analyses of zircon cores (Table 1), which yield ^{207}Pb*/^{206}Pb* dates (1σ) of 2739 ± 13 and 2691 ± 7 Ma.

Group P comprises five analyses (Table 1), which yield a ^{207}Pb*/^{206}Pb* dates of 2562–2508 Ma.

The date of 2650 ± 6 Ma for the six analyses in Group I is interpreted as the igneous crystallization age of the dacite. The dates (1σ) of 2739 ± 13 and 2691 ± 7 Ma for the two analyses of zircon cores in Group X are interpreted as the ages of inherited zircons. The dates of 2562–2508 Ma for the five analyses in Group P are interpreted to reflect loss of radiogenic Pb.

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Recommended reference for this publication

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